

## A9

December 5, 2025

```
[4]: pip install bitarray
```

```
Defaulting to user installation because normal site-packages is not writeable
Looking in links: /usr/share/pip-wheels
Requirement already satisfied: bitarray in ~/.local/lib/python3.11/site-packages
(3.8.0)
Note: you may need to restart the kernel to use updated packages.
```

```
[3]: pip install mmh3
```

```
Defaulting to user installation because normal site-packages is not writeable
Looking in links: /usr/share/pip-wheels
Collecting mmh3
  Obtaining dependency information for mmh3 from https://files.pythonhosted.org/
packages/00/7d/b7133b10d12239aeaebf6878d7eaf0bf7d3738c44b4aba3c564588f6d802/mmh3
-5.2.0-cp311-cp311-
manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl.metadata
  Downloading mmh3-5.2.0-cp311-cp311-
manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl.metadata (14
kB)
  Downloading mmh3-5.2.0-cp311-cp311-
manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl (103 kB)
103.1/103.1
kB 1.2 MB/s eta 0:00:00:00:0136m0:00:01
Installing collected packages: mmh3
Successfully installed mmh3-5.2.0
Note: you may need to restart the kernel to use updated packages.
```

```
[5]: pip install bloomf
```

```
Defaulting to user installation because normal site-packages is not writeable
Looking in links: /usr/share/pip-wheels
Collecting bloomf
  Obtaining dependency information for bloomf from https://files.pythonhosted.or
g/packages/88/8f/408cf8f074034f91e3f40512e8abb2fb694ae73b886f201487875715b139/bl
oomf-0.3-py3-none-any.whl.metadata
  Downloading bloomf-0.3-py3-none-any.whl.metadata (2.4 kB)
  Downloading bloomf-0.3-py3-none-any.whl (3.7 kB)
```

Installing collected packages: bloomf  
Successfully installed bloomf-0.3  
Note: you may need to restart the kernel to use updated packages.

```
[4]: pip install bloomfilter
```

Defaulting to user installation because normal site-packages is not writeable  
Looking in links: /usr/share/pip-wheels  
Collecting bloomfilter  
 Downloading BloomFilter-0.1.0.tar.gz (1.4 kB)  
 Preparing metadata (setup.py) ... done  
Building wheels for collected packages: bloomfilter  
 Building wheel for bloomfilter (setup.py) ... done  
 Created wheel for bloomfilter: filename=BloomFilter-0.1.0-py3-none-any.whl size=1615  
sha256=26ee70fd96c84718e0f034765d8a68f93a3854bf7aa3400c0d3a602c258308e8  
 Stored in directory: /home/c821c172-eb2e-4c76-bda4-16056937fc04/.cache/pip/wheels/d0/30/05/7d12ec2dbc016707103dc9a0aa74dfdb0c443407c738b4514c  
Successfully built bloomfilter  
Installing collected packages: bloomfilter  
Successfully installed bloomfilter-0.1.0  
Note: you may need to restart the kernel to use updated packages.

```
[10]: import mmh3  
from bitarray import bitarray  
import math
```

```
[14]: class BloomFilter:  
    def __init__(self, size, hash_count):  
        self.size = size  
        self.hash_count = hash_count  
        self.bit_array = bitarray(self.size)  
        self.bit_array.setall(0)  
    def _get_hashes(self, item):  
        hashes = []  
        for seed in range(self.hash_count):  
            h = mmh3.hash(item, seed) % self.size  
            hashes.append(h)  
        return hashes  
    def add(self, item):  
        digests = []  
        for i in self._get_hashes(item):  
            digest = mmh3.hash(item, i) % self.size  
            digests.append(digest)  
            self.bit_array[digest] = True  
    def __contains__(self, item):
```

```
    for i in self._get_hashes(item):
        digest = mmh3.hash(item, i) % self.size
        if self.bit_array[digest] == False:
            return False
    return True
```

```
[15]: blfilter = BloomFilter(size=1000, hash_count=5)
```

```
blfilter.add("UNT")
blfilter.add("compsci")

print("UNT" in blfilter)
print("compsci" in blfilter)
print("tigermascot" in blfilter)
```

```
True
True
False
```